

BCA 367

DSE V Lab — Programming in Go and IoT

T.Y. B.C.A. (Science) — Semester VI

CBCS 2019 Pattern | Slips 1–20 | All Q1 Options + IoT Appendix

MAEER's MIT Arts, Commerce and Science College, Alandi (D), Pune

Duration: 3 Hours | Max. Marks: 50

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Slip 1

BCA 367: Programming in Go and IoT | TY BCA (Science) Sem-VI | Max Marks: 50

Q1 (A) — Go Program

Write a program in GO to accept user choice and print answers using arithmetic operators. [20 Marks]

Program Code (Go):

main.go

```
package main
import "fmt"
func main() {
    var a, b float64
    var ch int
    fmt.Print("Enter two numbers: ")
    fmt.Scan(&a, &b)
    fmt.Println("1.Add 2.Sub 3.Mul 4.Div")
    fmt.Print("Enter choice: ")
    fmt.Scan(&ch)
    switch ch {
    case 1:
        fmt.Printf("Addition = %.2f\n", a+b)
    case 2:
        fmt.Printf("Subtraction = %.2f\n", a-b)
    case 3:
        fmt.Printf("Multiplication = %.2f\n", a*b)
    case 4:
        if b == 0 {
            fmt.Println("Division by zero!")
        } else {
            fmt.Printf("Division = %.2f\n", a/b)
        }
    default:
        fmt.Println("Invalid choice")
    }
}
```

Expected Output:

```
Enter two numbers: 8 5
1.Add 2.Sub 3.Mul 4.Div
Enter choice: 1
Addition = 13.00
```

OR

Q1 (B) — OR Go Program

Write a program in GO to accept n student details (roll_no, name, mark1, mark2, mark3). Calculate total and average using structure. [20 Marks]

Program Code (Go):

main.go

```
package main
import "fmt"
type Student struct {
    RollNo int
    Name string
    Mark1 float64
    Mark2 float64
    Mark3 float64
}
func main() {
    var n int
    fmt.Print("Enter number of students: ")
    fmt.Scan(&n)
    students := make([]Student, n)
```

```
for i := 0; i < n; i++ {  
    fmt.Printf("Enter rollno, name, m1, m2, m3: ")  
    fmt.Scan(&students[i].RollNo, &students[i].Name,  
            &students[i].Mark1, &students[i].Mark2, &students[i].Mark3)  
}  
for _, s := range students {  
    total := s.Mark1 + s.Mark2 + s.Mark3  
    avg := total / 3  
    fmt.Printf("Roll: %d | %s | Total: %.2f | Avg: %.2f\n",  
            s.RollNo, s.Name, total, avg)  
}  
}
```

Expected Output:

```
Enter number of students: 1  
Enter rollno, name, m1, m2, m3: 1 Rahul 80 75 90  
Roll: 1 | Rahul | Total: 245.00 | Avg: 81.67
```

Q2 — IoT Practical

Task: Blink LED (Prog 1) — see IoT Appendix at end of document

Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

Slip 2

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Q1 (A) — Go Program

Write a program in GO to print Fibonacci series of n terms. [20 Marks]

Program Code (Go):

```
main.go
package main
import "fmt"
func main() {
    var n int
    fmt.Print("Enter number of terms: ")
    fmt.Scan(&n)
    a, b := 0, 1
    fmt.Print("Fibonacci: ")
    for i := 0; i < n; i++ {
        fmt.Print(a, " ")
        a, b = b, a+b
    }
    fmt.Println()
}
```

Expected Output:

```
Enter number of terms: 8
Fibonacci: 0 1 1 2 3 5 8 13
```

OR

Q1 (B) — OR Go Program

Write a program in GO to print file information. [20 Marks]

Program Code (Go):

```
main.go
package main
import (
    "fmt"
    "os"
)
func main() {
    var filename string
    fmt.Print("Enter filename: ")
    fmt.Scan(&filename)
    info, err := os.Stat(filename)
    if err != nil {
        fmt.Println("Error:", err)
        return
    }
    fmt.Println("Name      :", info.Name())
    fmt.Println("Size       :", info.Size(), "bytes")
    fmt.Println("IsDir      :", info.IsDir())
    fmt.Println("Mode       :", info.Mode())
    fmt.Println("Modified:", info.ModTime())
}
```

Expected Output:

```
Enter filename: myfile.txt
Name       : myfile.txt
Size       : 28 bytes
IsDir      : false
Mode       : -rw-r--r--
Modified: 2026-03-22
```

Q2 — IoT Practical

Task: Buzzer ON/OFF (Prog 3) — see IoT Appendix at end of document

Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

Slip 3

BCA 367: Programming in Go and IoT | TY BCA (Science) Sem-VI | Max Marks: 50

Q1 (A) — Go Program

Write a program in GO using function to check whether accepted number is palindrome or not. **[20 Marks]**

Program Code (Go):

main.go

```
package main
import "fmt"
func isPalindrome(s string) bool {
    reversed := ""
    for i := len(s) - 1; i >= 0; i-- {
        reversed += string(s[i])
    }
    return s == reversed
}
func main() {
    str := "madam"
    if isPalindrome(str) {
        fmt.Println("Palindrome")
    } else {
        fmt.Println("Not Palindrome")
    }
}
```

Expected Output:

Palindrome

OR

Q1 (B) — OR Go Program

Write a program in GO to accept n employee records and display the employee with maximum salary. **[20 Marks]**

Program Code (Go):

main.go

```
package main
import "fmt"
type Employee struct {
    Eno    int
    Ename  string
    Salary float64
}
func main() {
    var n int
    fmt.Print("Enter number of employees: ")
    fmt.Scan(&n)
    emps := make([]Employee, n)
    for i := 0; i < n; i++ {
        fmt.Printf("Enter eno, ename, salary: ")
        fmt.Scan(&emps[i].Eno, &emps[i].Ename, &emps[i].Salary)
    }
    max := emps[0]
    for _, e := range emps[1:] {
        if e.Salary > max.Salary {
            max = e
        }
    }
    fmt.Printf("Max Salary: Eno=%d Name=%s Salary=%.2f\n",
        max.Eno, max.Ename, max.Salary)
}
```

Expected Output:

Enter number of employees: 3

...

Max Salary: Eno=2 Name=Priya Salary=75000.00

Q2 — IoT Practical

Task: Blink LED (Prog 1) — see IoT Appendix at end of document

Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

Slip 4

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Q1 (A) — Go Program

Write a program in GO to print recursive sum of digits of a given number. [20 Marks]

Program Code (Go):

main.go

```
package main
import "fmt"
func digitSum(n int) int {
    if n == 0 {
        return 0
    }
    return n%10 + digitSum(n/10)
}
func main() {
    var n int
    fmt.Print("Enter a number: ")
    fmt.Scan(&n)
    fmt.Printf("Sum of digits of %d = %d\n", n, digitSum(n))
}
```

Expected Output:

```
Enter a number: 1234
Sum of digits of 1234 = 10
```

OR

Q1 (B) — OR Go Program

Write a program in GO to sort array elements in ascending order. [20 Marks]

Program Code (Go):

main.go

```
package main
import "fmt"
func main() {
    var n int
    fmt.Print("Enter number of elements: ")
    fmt.Scan(&n)
    arr := make([]int, n)
    fmt.Println("Enter elements:")
    for i := 0; i < n; i++ {
        fmt.Scan(&arr[i])
    }
    for i := 0; i < n-1; i++ {
        for j := 0; j < n-i-1; j++ {
            if arr[j] > arr[j+1] {
                arr[j], arr[j+1] = arr[j+1], arr[j]
            }
        }
    }
    fmt.Println("Sorted:", arr)
}
```

Expected Output:

```
Enter number of elements: 5
Enter elements:
5 3 8 1 4
Sorted: [1 3 4 5 8]
```

Q2 — IoT Practical

Task: Toggle 2 LEDs (Prog 2) — see IoT Appendix at end of document

Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

Slip 5

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Q1 (A) — Go Program

Write a program in GO to create a text file. [20 Marks]

Program Code (Go):

main.go

```
package main
import (
    "fmt"
    "os"
)
func main() {
    file, err := os.Create("myfile.txt")
    if err != nil {
        fmt.Println("Error:", err)
        return
    }
    defer file.Close()
    file.WriteString("Hello, Go!\n")
    fmt.Println("File created successfully!")
}
```

Expected Output:

File created successfully!

OR

Q1 (B) — OR Go Program

Write a program in GO to accept n employee records and display the employee with minimum salary. [20 Marks]

Program Code (Go):

main.go

```
package main
import "fmt"
type Employee struct {
    Eno    int
    Ename  string
    Salary float64
}
func main() {
    var n int
    fmt.Print("Enter number of employees: ")
    fmt.Scan(&n)
    emps := make([]Employee, n)
    for i := 0; i < n; i++ {
        fmt.Printf("Enter eno, ename, salary: ")
        fmt.Scan(&emps[i].Eno, &emps[i].Ename, &emps[i].Salary)
    }
    min := emps[0]
    for _, e := range emps[1:] {
        if e.Salary < min.Salary {
            min = e
        }
    }
    fmt.Printf("Min Salary: Eno=%d Name=%s Salary=%.2f\n",
        min.Eno, min.Ename, min.Salary)
}
```

Expected Output:

Enter number of employees: 3
...

Min Salary: Eno=3 Name=Amit Salary=30000.00

Q2 — IoT Practical

Task: Blink LED (Prog 1) — see IoT Appendix at end of document

Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

Slip 6

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Q1 (A) — Go Program

Write a program in GO to accept two matrices and display their multiplication. [20 Marks]

Program Code (Go):

main.go

```
package main
import "fmt"
func main() {
    var r1, c1, r2, c2 int
    fmt.Print("Rows and cols of A: ")
    fmt.Scan(&r1, &c1)
    fmt.Print("Rows and cols of B: ")
    fmt.Scan(&r2, &c2)
    if c1 != r2 {
        fmt.Println("Multiplication not possible!")
        return
    }
    A := make([][]int, r1)
    B := make([][]int, r2)
    C := make([][]int, r1)
    for i := range A {
        A[i] = make([]int, c1)
    }
    for i := range B {
        B[i] = make([]int, c2)
    }
    for i := range C {
        C[i] = make([]int, c2)
    }
    fmt.Println("Enter Matrix A:")
    for i := 0; i < r1; i++ {
        for j := 0; j < c1; j++ {
            fmt.Scan(&A[i][j])
        }
    }
    fmt.Println("Enter Matrix B:")
    for i := 0; i < r2; i++ {
        for j := 0; j < c2; j++ {
            fmt.Scan(&B[i][j])
        }
    }
    for i := 0; i < r1; i++ {
        for j := 0; j < c2; j++ {
            for k := 0; k < c1; k++ {
                C[i][j] += A[i][k] * B[k][j]
            }
        }
    }
    fmt.Println("Result C = A x B:")
    for _, row := range C {
        fmt.Println(row)
    }
}
```

Expected Output:

```
Rows and cols of A: 2 2
Rows and cols of B: 2 2
Enter Matrix A:
1 2
3 4
Enter Matrix B:
```

```
5 6
7 8
Result C = A x B:
[19 22]
[43 50]
```

OR

Q1 (B) — OR Go Program

Write a program in GO to copy all elements of one array into another using a method. **[20 Marks]**

Program Code (Go):

```
main.go
package main
import "fmt"
func copyArr(src []int) []int {
    dst := make([]int, len(src))
    copy(dst, src)
    return dst
}
func main() {
    var n int
    fmt.Print("Enter number of elements: ")
    fmt.Scan(&n)
    arr := make([]int, n)
    fmt.Println("Enter elements:")
    for i := 0; i < n; i++ {
        fmt.Scan(&arr[i])
    }
    copied := copyArr(arr)
    fmt.Println("Original:", arr)
    fmt.Println("Copied  :", copied)
}
```

Expected Output:

```
Enter number of elements: 4
Enter elements:
10 20 30 40
Original: [10 20 30 40]
Copied  : [10 20 30 40]
```

Q2 — IoT Practical

Task: Toggle 2 LEDs (Prog 2) — see IoT Appendix at end of document

Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

Slip 7

BCA 367: Programming in Go and IoT | TY BCA (Science) Sem-VI | Max Marks: 50

Q1 (A) — Go Program

Write a program in GO to accept a matrix and display its transpose. **[20 Marks]**

Program Code (Go):

main.go

```
package main
import "fmt"
func main() {
    var r, c int
    fmt.Print("Enter rows and cols: ")
    fmt.Scan(&r, &c)
    A := make([][]int, r)
    for i := range A {
        A[i] = make([]int, c)
    }
    fmt.Println("Enter matrix:")
    for i := 0; i < r; i++ {
        for j := 0; j < c; j++ {
            fmt.Scan(&A[i][j])
        }
    }
    T := make([][]int, c)
    for i := range T {
        T[i] = make([]int, r)
    }
    for i := 0; i < r; i++ {
        for j := 0; j < c; j++ {
            T[j][i] = A[i][j]
        }
    }
    fmt.Println("Original:")
    for _, row := range A {
        fmt.Println(row)
    }
    fmt.Println("Transposed:")
    for _, row := range T {
        fmt.Println(row)
    }
}
```

Expected Output:

```
Enter rows and cols: 2 3
Enter matrix:
1 2 3
4 5 6
Original:
[1 2 3]
[4 5 6]
Transposed:
[1 4]
[2 5]
[3 6]
```

OR

Q1 (B) — OR Go Program

Write a program in GO to create structure Student. Write a method show() whose receiver is a pointer of struct Student. **[20 Marks]**

Program Code (Go):

main.go

```

package main
import "fmt"
type Student struct {
    RollNo int
    Name    string
    Marks   float64
}
func (s *Student) show() {
    fmt.Printf("Roll: %d | Name: %s | Marks: %.2f\n",
        s.RollNo, s.Name, s.Marks)
}
func main() {
    var n int
    fmt.Print("Enter number of students: ")
    fmt.Scan(&n)
    students := make([]Student, n)
    for i := 0; i < n; i++ {
        fmt.Printf("Enter rollno, name, marks: ")
        fmt.Scan(&students[i].RollNo, &students[i].Name, &students[i].Marks)
    }
    for i := range students {
        students[i].show()
    }
}

```

Expected Output:

```

Enter number of students: 1
Enter rollno, name, marks: 101 Rahul 85.5
Roll: 101 | Name: Rahul | Marks: 85.50

```

Q2 — IoT Practical

Task: Buzzer ON/OFF (Prog 3) — see IoT Appendix at end of document

Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

Slip 8

BCA 367: Programming in Go and IoT | TY BCA (Science) Sem-VI | Max Marks: 50

Q1 (A) — Go Program

Write a program in GO to accept n book details (BookID, Title, Author, Price) and display them. [20 Marks]

Program Code (Go):

```
main.go
package main
import "fmt"
type Book struct {
    BookID int
    Title  string
    Author string
    Price  float64
}
func main() {
    var n int
    fmt.Print("Enter number of books: ")
    fmt.Scan(&n)
    books := make([]Book, n)
    for i := 0; i < n; i++ {
        fmt.Printf("Enter BookID, Title, Author, Price: ")
        fmt.Scan(&books[i].BookID, &books[i].Title,
            &books[i].Author, &books[i].Price)
    }
    for _, b := range books {
        fmt.Printf("ID:%d | %s | %s | %.2f\n",
            b.BookID, b.Title, b.Author, b.Price)
    }
}
```

Expected Output:

```
Enter number of books: 1
Enter BookID, Title, Author, Price: 101 GoLang ABC 450
ID:101 | GoLang | ABC | 450.00
```

OR

Q1 (B) — OR Go Program

Write a program in GO to create an interface Shape with Area() and Perimeter(). Implement for Circle and Rectangle. [20 Marks]

Program Code (Go):

```
main.go
package main
import (
    "fmt"
    "math"
)
type Shape interface {
    Area() float64
    Perimeter() float64
}
type Circle struct {
    Radius float64
}
type Rectangle struct {
    Length, Width float64
}
func (c Circle) Area() float64 {
    return math.Pi * c.Radius * c.Radius
}
```

```

func (c Circle) Perimeter() float64 {
    return 2 * math.Pi * c.Radius
}
func (r Rectangle) Area() float64 {
    return r.Length * r.Width
}
func (r Rectangle) Perimeter() float64 {
    return 2 * (r.Length + r.Width)
}
func printShape(s Shape) {
    fmt.Printf("Area      = %.2f\n", s.Area())
    fmt.Printf("Perimeter = %.2f\n", s.Perimeter())
}
func main() {
    c := Circle{Radius: 5}
    r := Rectangle{Length: 4, Width: 3}
    fmt.Println("Circle:")
    printShape(c)
    fmt.Println("Rectangle:")
    printShape(r)
}

```

Expected Output:

```

Circle:
Area      = 78.54
Perimeter = 31.42
Rectangle:
Area      = 12.00
Perimeter = 14.00

```

Q2 — IoT Practical

Task: Blink LED (Prog 1) — see IoT Appendix at end of document

Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

Slip 9

BCA 367: Programming in Go and IoT | TY BCA (Science) Sem-VI | Max Marks: 50

Q1 (A) — Go Program

Write a program in GO using function to check whether accepted number is palindrome or not. **[20 Marks]**

Program Code (Go):

main.go

```
package main
import "fmt"
func isPalindrome(s string) bool {
    reversed := ""
    for i := len(s) - 1; i >= 0; i-- {
        reversed += string(s[i])
    }
    return s == reversed
}
func main() {
    str := "madam"
    if isPalindrome(str) {
        fmt.Println("Palindrome")
    } else {
        fmt.Println("Not Palindrome")
    }
}
```

Expected Output:

Palindrome

OR

Q1 (B) — OR Go Program

Write a program in GO to create an interface Shape with Area() and Volume(). Implement for Square and Rectangle. **[20 Marks]**

Program Code (Go):

main.go

```
package main
import "fmt"
type Shape interface {
    Area() float64
    Volume() float64
}
type Square struct {
    Side float64
}
type Rectangle struct {
    Length, Width float64
}
func (s Square) Area() float64 {
    return s.Side * s.Side
}
func (s Square) Volume() float64 {
    return s.Side * s.Side * s.Side
}
func (r Rectangle) Area() float64 {
    return r.Length * r.Width
}
func (r Rectangle) Volume() float64 {
    return r.Length * r.Width * r.Length
}
func main() {
    sq := Square{Side: 4}
```

```
rect := Rectangle{Length: 5, Width: 3}
fmt.Printf("Square:    Area=%.2f  Volume=%.2f\n", sq.Area(), sq.Volume())
fmt.Printf("Rectangle: Area=%.2f  Volume=%.2f\n", rect.Area(), rect.Volume())
}
```

Expected Output:

```
Square:    Area=16.00  Volume=64.00
Rectangle: Area=15.00  Volume=75.00
```

Q2 — IoT Practical

Task: Blink LED (Prog 1) — see IoT Appendix at end of document

Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

Slip 10

BCA 367: Programming in Go and IoT | TY BCA (Science) Sem-VI | Max Marks: 50

Q1 (A) — Go Program

Write a program in GO to create an interface and display its values with the help of type assertion.
[20 Marks]

Program Code (Go):

main.go

```
package main
import "fmt"
func printType(i interface{}) {
    switch v := i.(type) {
    case int:
        fmt.Printf("int      : %d\n", v)
    case float64:
        fmt.Printf("float64: %.2f\n", v)
    case string:
        fmt.Printf("string : %s\n", v)
    case bool:
        fmt.Printf("bool   : %v\n", v)
    default:
        fmt.Printf("unknown: %T\n", v)
    }
}
func main() {
    values := []interface{}{42, 3.14, "Hello Go", true}
    for _, val := range values {
        printType(val)
    }
}
```

Expected Output:

```
int      : 42
float64: 3.14
string : Hello Go
bool   : true
```

OR

Q1 (B) — OR Go Program

Write a program in GO to read and write Fibonacci series to a channel. [20 Marks]

Program Code (Go):

main.go

```
package main
import "fmt"
func fibonacci(n int, ch chan int) {
    a, b := 0, 1
    for i := 0; i < n; i++ {
        ch <- a
        a, b = b, a+b
    }
    close(ch)
}
func main() {
    var n int
    fmt.Print("Enter terms: ")
    fmt.Scan(&n)
    ch := make(chan int)
    go fibonacci(n, ch)
    fmt.Print("Fibonacci: ")
    for val := range ch {
        fmt.Print(val, " ")
    }
}
```

```
}  
    fmt.Println()  
}
```

Expected Output:

```
Enter terms: 8  
Fibonacci: 0 1 1 2 3 5 8 13
```

Q2 — IoT Practical

Task: Buzzer ON/OFF (Prog 3) — see IoT Appendix at end of document

Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

Slip 11

BCA 367: Programming in Go and IoT | TY BCA (Science) Sem-VI | Max Marks: 50

Q1 (A) — Go Program

Write a program in GO to check whether the accepted number is a two-digit number or not. [20 Marks]

Program Code (Go):

```
main.go
package main
import "fmt"
func isTwoDigit(n int) bool {
    if n < 0 {
        n = -n
    }
    return n >= 10 && n <= 99
}
func main() {
    var n int
    fmt.Print("Enter a number: ")
    fmt.Scan(&n)
    if isTwoDigit(n) {
        fmt.Printf("%d is a Two-Digit number\n", n)
    } else {
        fmt.Printf("%d is NOT a Two-Digit number\n", n)
    }
}
```

Expected Output:

```
Enter a number: 45
45 is a Two-Digit number
---
Enter a number: 100
100 is NOT a Two-Digit number
```

OR

Q1 (B) — OR Go Program

Write a program in GO to create a buffered channel, store values in it and find capacity and length. [20 Marks]

Program Code (Go):

```
main.go
package main
import "fmt"
func main() {
    ch := make(chan int, 5)
    ch <- 10
    ch <- 20
    ch <- 30
    fmt.Println("Capacity:", cap(ch))
    fmt.Println("Length :", len(ch))
    v1 := <-ch
    v2 := <-ch
    fmt.Println("Read      :", v1, v2)
    fmt.Println("New Len  :", len(ch))
}
```

Expected Output:

```
Capacity: 5
Length  : 3
Read    : 10 20
New Len : 1
```

Q2 — IoT Practical

Task: Buzzer ON/OFF (Prog 3) — see IoT Appendix at end of document

Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

Slip 12

BCA 367: Programming in Go and IoT | TY BCA (Science) Sem-VI | Max Marks: 50

Q1 (A) — Go Program

Write a program in GO to swap two numbers using call by reference. **[20 Marks]**

Program Code (Go):

main.go

```
package main
import "fmt"
func swap(a, b *int) {
    *a, *b = *b, *a
}
func main() {
    x, y := 10, 20
    fmt.Printf("Before: x=%d y=%d\n", x, y)
    swap(&x, &y)
    fmt.Printf("After : x=%d y=%d\n", x, y)
}
```

Expected Output:

```
Before: x=10 y=20
After : x=20 y=10
```

OR

Q1 (B) — OR Go Program

Write a program in GO that creates a slice of integers, checks for even/odd and sends to respective goroutines via channel. **[20 Marks]**

Program Code (Go):

main.go

```
package main
import (
    "fmt"
    "sync"
)
func printEven(nums []int, wg *sync.WaitGroup) {
    defer wg.Done()
    fmt.Print("Even: ")
    for _, n := range nums {
        if n%2 == 0 {
            fmt.Print(n, " ")
        }
    }
    fmt.Println()
}
func printOdd(nums []int, wg *sync.WaitGroup) {
    defer wg.Done()
    fmt.Print("Odd : ")
    for _, n := range nums {
        if n%2 != 0 {
            fmt.Print(n, " ")
        }
    }
    fmt.Println()
}
func main() {
    nums := []int{1, 2, 3, 4, 5, 6, 7, 8, 9, 10}
    var wg sync.WaitGroup
    wg.Add(2)
    go printEven(nums, &wg)
    go printOdd(nums, &wg)
    wg.Wait()
}
```

```
}
```

Expected Output:

Even: 2 4 6 8 10

Odd : 1 3 5 7 9

Q2 — IoT Practical

Task: Toggle 2 LEDs (Prog 2) — see IoT Appendix at end of document

Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

Slip 13

BCA 367: Programming in Go and IoT | TY BCA (Science) Sem-VI | Max Marks: 50

Q1 (A) — Go Program

Write a program in GO to print the sum of even and odd numbers separately between 1 to 100. [20 Marks]

Program Code (Go):

```
main.go
package main
import "fmt"
func main() {
    evenSum, oddSum := 0, 0
    for i := 1; i <= 100; i++ {
        if i%2 == 0 {
            evenSum += i
        } else {
            oddSum += i
        }
    }
    fmt.Println("Sum of Even (1-100):", evenSum)
    fmt.Println("Sum of Odd (1-100):", oddSum)
}
```

Expected Output:

```
Sum of Even (1-100): 2550
Sum of Odd (1-100): 2500
```

OR

Q1 (B) — OR Go Program

Write a function in GO to find the square of a number and write a benchmark for it. [20 Marks]

Program Code (Go):

```
main.go
package main
import (
    "fmt"
    "time"
)
func square(n int) int {
    return n * n
}
func main() {
    var n int
    fmt.Print("Enter a number: ")
    fmt.Scan(&n)
    fmt.Printf("Square of %d = %d\n", n, square(n))
    start := time.Now()
    for i := 0; i < 1000000; i++ {
        square(i)
    }
    fmt.Printf("Benchmark: 1M calls took %v\n", time.Since(start))
}
```

Expected Output:

```
Enter a number: 7
Square of 7 = 49
Benchmark: 1M calls took ~1.5ms
```

Q2 — IoT Practical

Task: Toggle 2 LEDs (Prog 2) — see IoT Appendix at end of document

Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

Slip 14

BCA 367: Programming in Go and IoT | TY BCA (Science) Sem-VI | Max Marks: 50

Q1 (A) — Go Program

Write a program in GO to demonstrate working of slices (append, remove, copy etc.). **[20 Marks]**

Program Code (Go):

main.go

```
package main
import "fmt"
func main() {
    s := []int{10, 20, 30, 40, 50}
    fmt.Println("Original:", s)
    s = append(s, 60, 70)
    fmt.Println("Append :", s)
    s2 := make([]int, len(s))
    copy(s2, s)
    fmt.Println("Copy   :", s2)
    s = append(s[:2], s[3:]...)
    fmt.Println("Remove :", s)
    fmt.Printf("Len=%d Cap=%d\n", len(s), cap(s))
}
```

Expected Output:

```
Original: [10 20 30 40 50]
Append  : [10 20 30 40 50 60 70]
Copy    : [10 20 30 40 50 60 70]
Remove  : [10 20 40 50 60 70]
```

OR

Q1 (B) — OR Go Program

Write a GO program using goroutine and channel to print sum of squares and cubes of individual digits of a number. **[20 Marks]**

Program Code (Go):

main.go

```
package main
import (
    "fmt"
    "sync"
)
func sumSquares(digits []int, ch chan<- int, wg *sync.WaitGroup) {
    defer wg.Done()
    sum := 0
    for _, d := range digits {
        sum += d * d
    }
    ch <- sum
}
func sumCubes(digits []int, ch chan<- int, wg *sync.WaitGroup) {
    defer wg.Done()
    sum := 0
    for _, d := range digits {
        sum += d * d * d
    }
    ch <- sum
}
func main() {
    var n int
    fmt.Print("Enter a number (e.g. 123): ")
    fmt.Scan(&n)
    digits := []int{}
    temp := n
```

```

    for temp > 0 {
        digits = append([]int{temp % 10}, digits...)
        temp /= 10
    }
    sqCh := make(chan int, 1)
    cuCh := make(chan int, 1)
    var wg sync.WaitGroup
    wg.Add(2)
    go sumSquares(digits, sqCh, &wg)
    go sumCubes(digits, cuCh, &wg)
    wg.Wait()
    close(sqCh)
    close(cuCh)
    sq, cu := <-sqCh, <-cuCh
    fmt.Printf("Sum of Squares = %d\nSum of Cubes    = %d\nFinal Sum        = %d\n",
        sq, cu, sq+cu)
}

```

Expected Output:

```

Enter a number (e.g. 123): 123
Sum of Squares = 14
Sum of Cubes   = 36
Final Sum      = 50

```

Q2 — IoT Practical

Task: Buzzer ON/OFF (Prog 3) — see IoT Appendix at end of document

Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

Slip 15

BCA 367: Programming in Go and IoT | TY BCA (Science) Sem-VI | Max Marks: 50

Q1 (A) — Go Program

Write a program in GO to demonstrate function returning multiple values (min and max). [20 Marks]

Program Code (Go):

main.go

```
package main
import "fmt"
func minMax(arr []int) (int, int) {
    min, max := arr[0], arr[0]
    for _, v := range arr[1:] {
        if v < min {
            min = v
        }
        if v > max {
            max = v
        }
    }
    return min, max
}
func main() {
    var n int
    fmt.Print("Enter number of elements: ")
    fmt.Scan(&n)
    arr := make([]int, n)
    fmt.Println("Enter elements:")
    for i := range arr {
        fmt.Scan(&arr[i])
    }
    min, max := minMax(arr)
    fmt.Printf("Minimum = %d\nMaximum = %d\n", min, max)
}
```

Expected Output:

```
Enter number of elements: 5
Enter elements:
3 7 1 9 4
Minimum = 1
Maximum = 9
```

OR

Q1 (B) — OR Go Program

Write a program in GO to read an XML file into a structure and display it. [20 Marks]

Program Code (Go):

main.go

```
package main
import (
    "encoding/xml"
    "fmt"
    "os"
)
type Note struct {
    To      string `xml:"to"`
    From    string `xml:"from"`
    Heading string `xml:"heading"`
    Body    string `xml:"body"`
}
func main() {
    data := "<note><to>Student</to><from>Teacher</from><heading>Assignment</heading><body>Submit by Friday!</body></note>"
```

```
os.WriteFile("notes.xml", []byte(data), 0644)
raw, err := os.ReadFile("notes.xml")
if err != nil {
    fmt.Println("Error:", err)
    return
}
var note Note
xml.Unmarshal(raw, &note)
fmt.Println("To      :", note.To)
fmt.Println("From    :", note.From)
fmt.Println("Heading:", note.Hheading)
fmt.Println("Body     :", note.Body)
}
```

Expected Output:

```
To      : Student
From    : Teacher
Heading: Assignment
Body     : Submit by Friday!
```

Q2 — IoT Practical

Task: Toggle 2 LEDs (Prog 2) — see IoT Appendix at end of document

Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

Slip 16

BCA 367: Programming in Go and IoT | TY BCA (Science) Sem-VI | Max Marks: 50

Q1 (A) — Go Program

Write a program in GO to create a user-defined package to find the area of a rectangle. **[20 Marks]**

Program Code (Go):

```
main.go
// File: rect/rect.go
package rect

func Area(l, w float64) float64 {
    return l * w
}
func Perimeter(l, w float64) float64 {
    return 2 * (l + w)
}

// File: main.go
package main
import (
    "fmt"
    "rect"
)
func main() {
    var l, w float64
    fmt.Print("Enter length and width: ")
    fmt.Scan(&l, &w)
    fmt.Printf("Area      = %.2f\n", rect.Area(l, w))
    fmt.Printf("Perimeter = %.2f\n", rect.Perimeter(l, w))
}
```

Expected Output:

```
Enter length and width: 5 3
Area      = 15.00
Perimeter = 16.00
```

OR

Q1 (B) — OR Go Program

Write a program in GO to print numbers 0 to 10 with a random delay of 0 to 250ms after each. **[20 Marks]**

Program Code (Go):

```
main.go
package main
import (
    "fmt"
    "math/rand"
    "time"
)
func main() {
    rand.Seed(time.Now().UnixNano())
    for i := 0; i <= 10; i++ {
        fmt.Printf("Number: %d\n", i)
        delay := time.Duration(rand.Intn(251)) * time.Millisecond
        time.Sleep(delay)
    }
}
```

Expected Output:

```
Number: 0
Number: 1
...
```

Number: 10
(each after random 0-250ms delay)

Q2 — IoT Practical

Task: Blink LED (Prog 1) — see IoT Appendix at end of document

Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

Slip 17

BCA 367: Programming in Go and IoT | TY BCA (Science) Sem-VI | Max Marks: 50

Q1 (A) — Go Program

Write a program in GO to return multiple values (Add, Subtract, Multiply, Divide) from a function. [20 Marks]

Program Code (Go):

```
main.go
package main
import "fmt"
func calculate(a, b float64) (float64, float64, float64, float64) {
    return a + b, a - b, a * b, a / b
}
func main() {
    var a, b float64
    fmt.Print("Enter two numbers: ")
    fmt.Scan(&a, &b)
    add, sub, mul, div := calculate(a, b)
    fmt.Printf("Add = %.2f\nSub = %.2f\nMul = %.2f\nDiv = %.2f\n",
        add, sub, mul, div)
}
```

Expected Output:

```
Enter two numbers: 10 3
Add = 13.00
Sub = 7.00
Mul = 30.00
Div = 3.33
```

OR

Q1 (B) — OR Go Program

Write a program in GO to add/append content at the end of a text file. [20 Marks]

Program Code (Go):

```
main.go
package main
import (
    "fmt"
    "os"
)
func main() {
    file, err := os.OpenFile("data.txt",
        os.O_APPEND|os.O_CREATE|os.O_WRONLY, 0644)
    if err != nil {
        fmt.Println("Error:", err)
        return
    }
    defer file.Close()
    var text string
    fmt.Print("Enter text to append: ")
    fmt.Scan(&text)
    file.WriteString(text + "\n")
    fmt.Println("Appended successfully!")
}
```

Expected Output:

```
Enter text to append: Hello World
Appended successfully!
```

Q2 — IoT Practical

Task: Toggle 2 LEDs (Prog 2) — see IoT Appendix at end of document

Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

Slip 18

BCA 367: Programming in Go and IoT | TY BCA (Science) Sem-VI | Max Marks: 50

Q1 (A) — Go Program

Write a program in GO to print a multiplication table of a number using a function. [20 Marks]

Program Code (Go):

main.go

```
package main
import "fmt"
func printTable(n int) {
    fmt.Printf("Table of %d:\n", n)
    for i := 1; i <= 10; i++ {
        fmt.Printf("%d x %d = %d\n", n, i, n*i)
    }
}
func main() {
    var n int
    fmt.Print("Enter a number: ")
    fmt.Scan(&n)
    printTable(n)
}
```

Expected Output:

```
Enter a number: 5
Table of 5:
5 x 1 = 5
5 x 2 = 10
...
5 x 10 = 50
```

OR

Q1 (B) — OR Go Program

Write a program in GO using a user-defined package calculator that performs arithmetic operations. [20 Marks]

Program Code (Go):

main.go

```
// File: calculator/calculator.go
package calculator

import "fmt"

func Calculate(a, b float64, ch int) {
    switch ch {
    case 1:
        fmt.Printf("Addition      = %.2f\n", a+b)
    case 2:
        fmt.Printf("Subtraction   = %.2f\n", a-b)
    case 3:
        fmt.Printf("Multiplication = %.2f\n", a*b)
    case 4:
        if b == 0 {
            fmt.Println("Division by zero!")
        } else {
            fmt.Printf("Division      = %.2f\n", a/b)
        }
    default:
        fmt.Println("Invalid choice")
    }
}
```

```
// File: main.go
package main
import (
    "calculator"
    "fmt"
)
func main() {
    var a, b float64
    var ch int
    fmt.Print("Enter two numbers: ")
    fmt.Scan(&a, &b)
    fmt.Println("1.Add 2.Sub 3.Mul 4.Div")
    fmt.Print("Enter choice: ")
    fmt.Scan(&ch)
    calculator.Calculate(a, b, ch)
}
```

Expected Output:

```
Enter two numbers: 8 4
1.Add 2.Sub 3.Mul 4.Div
Enter choice: 3
Multiplication = 32.00
```

Q2 — IoT Practical

Task: Buzzer ON/OFF (Prog 3) — see IoT Appendix at end of document

Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

Slip 19

BCA 367: Programming in Go and IoT | TY BCA (Science) Sem-VI | Max Marks: 50

Q1 (A) — Go Program

Write a program in GO to return multiple values (add, subtract) from a function. [20 Marks]

Program Code (Go):

main.go

```
package main
import "fmt"
func addSub(a, b int) (int, int) {
    return a + b, a - b
}
func main() {
    var a, b int
    fmt.Print("Enter two numbers: ")
    fmt.Scan(&a, &b)
    sum, diff := addSub(a, b)
    fmt.Printf("Sum          = %d\nDifference = %d\n", sum, diff)
}
```

Expected Output:

```
Enter two numbers: 15 6
Sum          = 21
Difference = 9
```

OR

Q1 (B) — OR Go Program

Write a program in GO to open a file in READ-ONLY mode. [20 Marks]

Program Code (Go):

main.go

```
package main
import (
    "bufio"
    "fmt"
    "os"
)
func main() {
    var filename string
    fmt.Print("Enter filename: ")
    fmt.Scan(&filename)
    file, err := os.OpenFile(filename, os.O_RDONLY, 0444)
    if err != nil {
        fmt.Println("Error:", err)
        return
    }
    defer file.Close()
    fmt.Println("Contents:")
    scanner := bufio.NewScanner(file)
    for scanner.Scan() {
        fmt.Println(scanner.Text())
    }
}
```

Expected Output:

```
Enter filename: data.txt
Contents:
Hello World
```

Q2 — IoT Practical

Task: Buzzer ON/OFF (Prog 3) — see IoT Appendix at end of document

Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

Slip 20

BCA 367: Programming in Go and IoT | TY BCA (Science) Sem-VI | Max Marks: 50

Q1 (A) — Go Program

Write a program in GO to add/append content at the end of a text file. **[20 Marks]**

Program Code (Go):

```
main.go
package main
import (
    "fmt"
    "os"
)
func main() {
    file, err := os.OpenFile("data.txt",
        os.O_APPEND|os.O_CREATE|os.O_WRONLY, 0644)
    if err != nil {
        fmt.Println("Error:", err)
        return
    }
    defer file.Close()
    var text string
    fmt.Print("Enter text to append: ")
    fmt.Scan(&text)
    file.WriteString(text + "\n")
    fmt.Println("Appended successfully!")
}
```

Expected Output:

```
Enter text to append: Go Programming
Appended successfully!
```

OR

Q1 (B) — OR Go Program

Write a program in GO to create a channel and demonstrate closing it using for-range loop. **[20 Marks]**

Program Code (Go):

```
main.go
package main
import "fmt"
func send(ch chan int) {
    for i := 1; i <= 5; i++ {
        ch <- i * 10
    }
    close(ch)
}
func main() {
    ch := make(chan int)
    go send(ch)
    for val := range ch {
        fmt.Println("Received:", val)
    }
    fmt.Println("Channel closed.")
}
```

Expected Output:

```
Received: 10
Received: 20
Received: 30
Received: 40
Received: 50
Channel closed.
```

Q2 — IoT Practical

Task: Toggle 2 LEDs (Prog 2) — see IoT Appendix at end of document

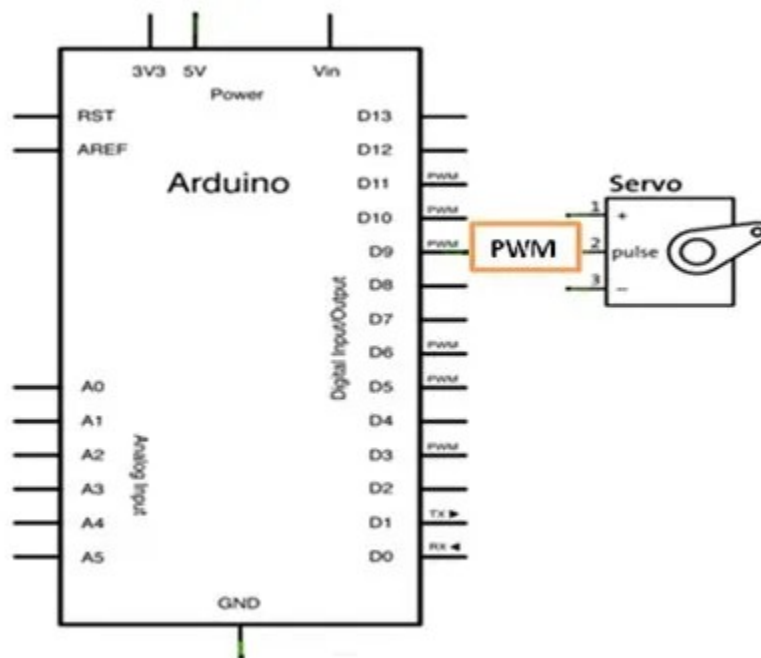
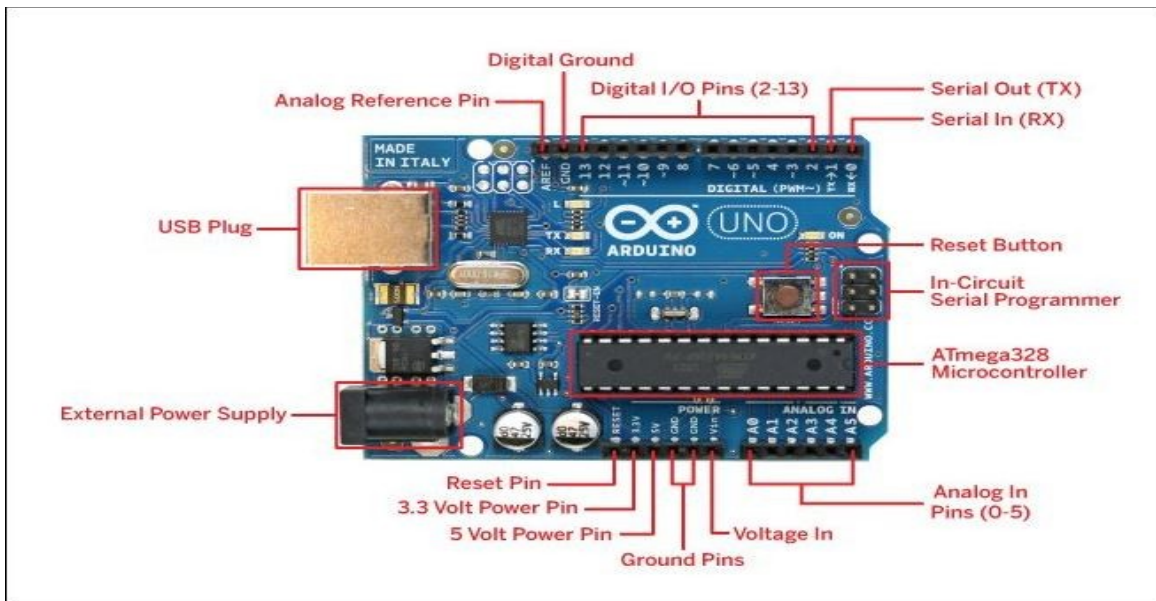
Q3. Viva [5 Marks] Q4. Internal Assessment [15 Marks]

APPENDIX — IoT Practical (Arduino UNO) BCA 367 Q2 — Common for all Slips 1–20

Q2 Question Format (same for all slips)

- Draw block diagram of Arduino Uno / Raspberry Pi / Beagle board interfacing with IR Sensor.
- WAP in C++ to blink LED OR toggle two LEDs OR turn ON/OFF buzzer.
- Write observations (Input / Output).
- Write Result and Conclusion.

Circuit / Block Diagram (from Lab Book)



Program 1 (C++): Blink LED — Used in Slips: 1, 3, 5, 8, 9, 15, 16

blink_led.ino

```
// Arduino UNO - Blink LED
// Red LED(+) -> Pin 12 -> 220 ohm resistor -> GND

const int LED = 12;

void setup() {
    pinMode(LED, OUTPUT);
}

void loop() {
    digitalWrite(LED, HIGH);
    delay(1000);
    digitalWrite(LED, LOW);
    delay(1000);
}
```

Expected Output:

LED turns ON for 1 second
LED turns OFF for 1 second
(Repeats indefinitely)

Observations (Input & Output)

After uploading the program, the LED connected to pin 12 starts blinking

LED turns **ON for 1 second** and **OFF for 1 second**

The blinking continues repeatedly in a loop

No external input is required (program runs automatically)

Result and Conclusion

Result:

The LED blinking program was executed successfully on Arduino Uno

Conclusion:

Arduino Uno can control output devices like LED using simple programming

The digital output pins can be used to turn devices ON and OFF

The experiment verifies basic working of Arduino and GPIO pins

Program 2 (C++): Toggle Two LEDs — Used in Slips: 4, 6, 12, 13, 14, 17, 20

toggle_leds.ino

```
// Arduino UNO - Toggle Two LEDs
// Red LED(+) -> Pin 12 -> 220 ohm -> GND
// Green LED(+) -> Pin 11 -> 220 ohm -> GND

const int RED = 12;
const int GREEN = 11;

void setup() {
    pinMode(RED, OUTPUT);
    pinMode(GREEN, OUTPUT);
}

void loop() {
```

```
digitalWrite(RED, HIGH);  
digitalWrite(GREEN, LOW);  
delay(1000);  
digitalWrite(RED, LOW);  
digitalWrite(GREEN, HIGH);  
delay(1000);  
}
```

Expected Output:

RED ON | GREEN OFF
RED OFF | GREEN ON
(Alternates every 1 second)

Observations (Input & Output)

After uploading the program, both LEDs connected to the Arduino start operating

LED 1 turns ON while LED 2 remains OFF

After 1 second, LED 1 turns OFF and LED 2 turns ON

The LEDs glow alternately (toggle)

The process continues repeatedly in a loop

No external input is required (program runs automatically)

Result and Conclusion

Result:

The two LED toggle program was executed successfully on Arduino Uno

Conclusion:

Arduino Uno can control multiple output devices simultaneously

Digital output pins can be used to toggle devices

The experiment verifies basic working of Arduino and GPIO pins

Program 3 (C++): Buzzer ON/OFF — Used in Slips: 2, 7, 10, 11, 18, 19

buzzer.ino

```
// Arduino UNO - Buzzer ON/OFF  
// Active Buzzer(+) -> Pin 9 -> GND  
  
const int BUZZER = 9;  
  
void setup() {  
    pinMode(BUZZER, OUTPUT);  
}  
  
void loop() {  
    digitalWrite(BUZZER, HIGH);  
    delay(1000);  
    digitalWrite(BUZZER, LOW);  
    delay(1000);  
}
```

Expected Output:

Buzzer ON for 1 second
Buzzer OFF for 1 second
(Repeats indefinitely)

Observations (Input & Output)

After uploading the program, the buzzer connected to pin 9 starts operating

Buzzer turns ON for 1 second and OFF for 1 second

Sound is produced when the buzzer is ON

The ON/OFF cycle continues repeatedly in a loop

No external input is required (program runs automatically)

Result and Conclusion

Result:

The buzzer ON/OFF program was executed successfully on Arduino Uno

Conclusion:

Arduino Uno can control output devices like buzzer using simple programming

The digital output pins can be used to turn devices ON and OFF

The experiment verifies basic working of Arduino and GPIO pins